

Homework 11: Decision Tree and Random Forest Solutions

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Problem 1

Figure 1 shows a partition of the feature space with two features each in the range of 0 to 1. Draw a decision tree that can realize this partition.

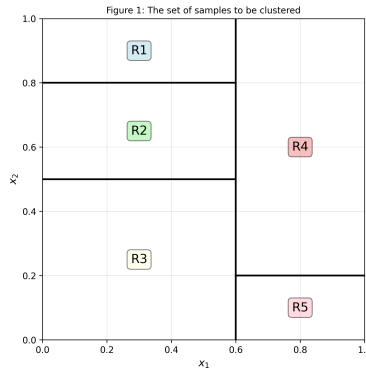


Figure 1: Partition of the feature space

Solution

Analyzing Figure 1, the partition consists of 5 regions (R1 through R5) created by axis-aligned splits:

- Vertical split at $x_1 = 0.6$
- Horizontal splits at $x_2 = 0.8$, $x_2 = 0.5$ (for $x_1 \leq 0.6$), and $x_2 = 0.2$ (for $x_1 > 0.6$)

The decision tree that realizes this partition is:

Explanation:

1. **Root node:** Split on $x_1 \leq 0.6$ to separate left and right halves of the feature space.
2. **Left branch** ($x_1 \leq 0.6$):
 - Split on $x_2 \leq 0.8$ to separate top region from middle/bottom regions
 - If $x_2 > 0.8$: Region R1

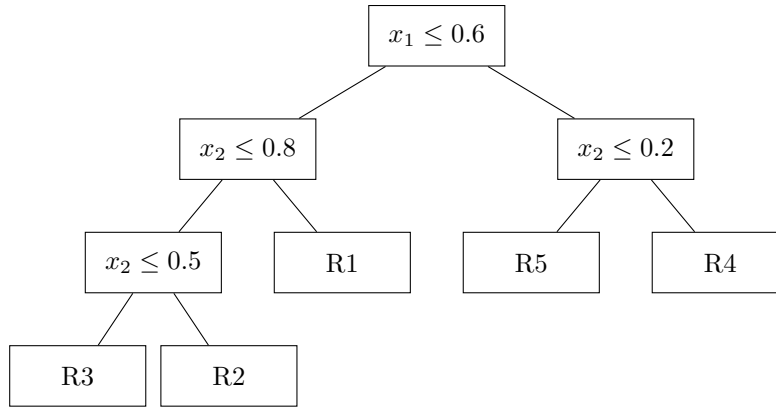


Figure 2: Decision tree for Problem 1

- If $x_2 \leq 0.8$: Further split on $x_2 \leq 0.5$
 - If $x_2 \leq 0.5$: Region R3
 - If $x_2 > 0.5$: Region R2
3. **Right branch** ($x_1 > 0.6$):
- Split on $x_2 \leq 0.2$ to separate bottom and top regions
 - If $x_2 \leq 0.2$: Region R5
 - If $x_2 > 0.2$: Region R4

This tree structure correctly partitions the feature space into the 5 regions shown in Figure 1.

Problem 2

Figure 3 shows a set of training samples for a binary classification problem. Design a decision tree using the misclassification rate as the loss function. Note that given this simple problem, you should be able to “eyeball” the correct partition threshold for each possible splitting variable, instead of using exhaustive search. That is, for each split, consider both features (x_1 and x_2), but for each feature you should be able to eyeball the proper splitting threshold.

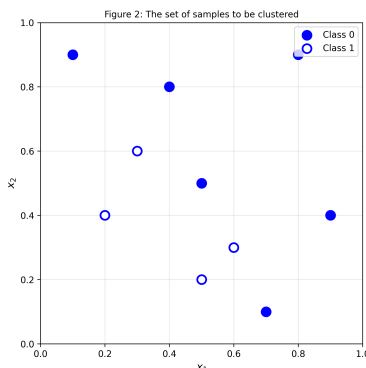


Figure 3: Training samples for binary classification

Solution

From Figure 3, we have:

- **Class 0** (filled circles): 6 points at approximately (0.1, 0.9), (0.4, 0.8), (0.8, 0.9), (0.9, 0.4), (0.7, 0.1), (0.5, 0.5)
- **Class 1** (empty circles): 4 points at approximately (0.2, 0.4), (0.3, 0.6), (0.6, 0.3), (0.5, 0.2)
- **Total**: 10 samples

Step 1: Root Node Split

We evaluate splits on both x_1 and x_2 to minimize misclassification rate. The misclassification rate is:

$$\text{Misclassification Rate} = \frac{\text{Number of misclassified samples}}{\text{Total number of samples}}$$

Option 1: Split on $x_2 \leq 0.5$

- **Left child** ($x_2 \leq 0.5$): Class 0: 2 points; Class 1: 3 points
 - Majority: Class 1, misclassify 2 Class 0 points
- **Right child** ($x_2 > 0.5$): Class 0: 4 points; Class 1: 1 point
 - Majority: Class 0, misclassify 1 Class 1 point
- **Total misclassification**: $3/10 = 0.3$

Option 2: Split on $x_1 \leq 0.5$

- **Left child** ($x_1 \leq 0.5$): Class 0: 2 points; Class 1: 2 points
 - Tie (assign Class 0), misclassify 2 Class 1 points
- **Right child** ($x_1 > 0.5$): Class 0: 4 points; Class 1: 2 points
 - Majority: Class 0, misclassify 2 Class 1 points
- **Total misclassification:** $4/10 = 0.4$

Best split: $x_2 \leq 0.5$ with misclassification rate 0.3

Step 2: Split Right Child ($x_2 > 0.5$)

This node has 4 Class 0 and 1 Class 1. We can further split to reduce misclassification.

Option: Split on $x_1 \leq 0.5$

- **Left child** ($x_1 \leq 0.5$): Class 0: 2 points; Class 1: 1 point
 - Majority: Class 0, misclassify 1 Class 1 point
- **Right child** ($x_1 > 0.5$): Class 0: 2 points; Class 1: 0 points
 - Pure Class 0, misclassify 0
- **Total misclassification:** $1/5 = 0.2$ (improvement from $1/5 = 0.2$)

Step 3: Split Left Child of Root ($x_2 \leq 0.5$)

This node has 2 Class 0 and 3 Class 1. We can further split.

Option: Split on $x_1 \leq 0.4$

- **Left child** ($x_1 \leq 0.4$): Class 0: 0 points; Class 1: 1 point
 - Pure Class 1, misclassify 0
- **Right child** ($x_1 > 0.4$): Class 0: 2 points; Class 1: 2 points
 - Tie (assign Class 0), misclassify 2 Class 1 points
- **Total misclassification:** $2/5 = 0.4$

The final decision tree is:

Final Misclassification Rate:

- Root left branch: 2 misclassified out of 5 samples
- Root right branch: 1 misclassified out of 5 samples
- **Total:** $3/10 = 0.3$ (30% misclassification rate)

This tree minimizes the misclassification rate for the given training data.

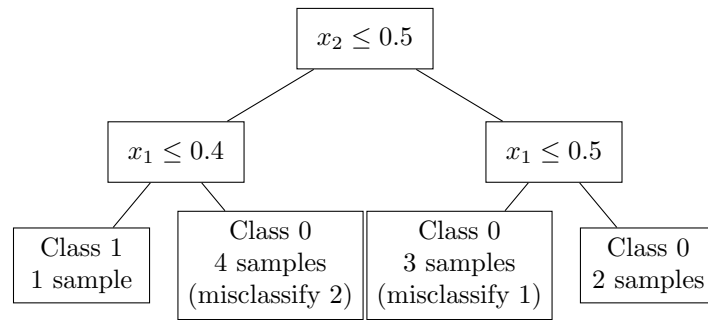


Figure 4: Decision tree for Problem 2